

DEPT: Decoupled Embeddings for Pretraining Language Models

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Vocabulary-agnostic, communication-efficient pretraining by decoupling token and positional embeddings from the transformer body.



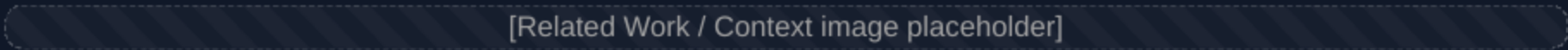
Abstract & Motivation

- Heterogeneous data in LM pretraining induces negative interference and the curse of multilinguality.
- They propose DEPT to decouple embeddings from the transformer to train across sources without a shared vocabulary.
- Reported benefits: robustness under heterogeneity; reduced embedding parameters and communication; improved perplexity (up to 20%) and downstream; custom per-source vocabularies.
- The approach enables vocabulary-agnostic federated pretraining of billion-scale models and 4–5× embedding memory reduction.



Introduction

- Scaling data increases heterogeneity across domains and languages.
- Key challenges include capacity contention, vocabulary dilution, suboptimal tokenization, and costly sampling/HPO.
- Prior evidence indicates custom vocabularies improve performance across languages/domains.
- The goal is to decouple embeddings so the transformer learns abstract, vocabulary-independent representations.



DEPT Pipeline

1. Build vocabulary/tokenizer (global or per-source).
2. Pre-tokenize corpora per source.
3. Workers train locally on pre-tokenized data.
4. Collect partial results; aggregate transformer body.
5. Broadcast updated transformer; repeat.



Method: Optimization Setup

- Optimization targets: global θ (transformer body) with optional ϕ (token) and ψ (positional) per source.
- Training iterates over a random subset S_t ; InnerOPT runs per source; OuterOPT aggregates bodies.
- Three variants are introduced for handling embeddings: GLOB, TRIM, SPEC.

Key idea: transformer body learning is largely embedding-independent; embeddings can be local/specialized.

Variants: GLOB · TRIM · SPEC

- GLOB

 - Shares and aggregates transformer plus embeddings (FedAvg-like).
 - Reduces communication via N_{local} -step synchronization.
 - Requires a global vocabulary and risks dilution.
- TRIM

 - Shares transformer with embeddings trimmed to local V_k .
 - Projects back with zero-padding to save memory and communication.
 - Restricts output space under tied weights.
- SPEC

 - Shares only the transformer body; local ϕ and ψ remain specialized and unaggregated.
 - Is vocabulary- and length-agnostic with minimal communication.
 - For inference, a global embedding is created via continued pretraining.
- [Variant schematic placeholder]

TRIM: Embedding Aggregation

- $\phi_k = I_k \phi$ selects rows for local V_k .
- After local training, projection uses $\hat{\phi}_k = I_k^T \phi_k$ with zero-padding of absent tokens.
- Outer aggregation ignores zeros to avoid cross-source interference.

[Projection matrix illustration placeholder]

SPEC: Positional Specialization

- Specializes both token (ϕ) and positional (ψ) embeddings.
- Supports disjoint per-source vocabularies and sequence lengths.
- Training is vocabulary-agnostic; embeddings are created later via continued pretraining.

[SPEC conceptual placeholder]

Memory & Communication Costs

- STD: $O(M)$ memory; per-step communication $O(M)$.
- GLOB: $O(M)$ memory; communication every N_{local} steps.
- TRIM: saves $(|V|-|V_k|) \cdot d_{model}$ parameters; reduced communication.
- SPEC: has no embedding communication, is vocab-agnostic, and yields additional ψ savings.

Observed: up to 714× communication reduction (1B params) and 4–5× embedding memory savings.

Experimental Design

- Models: decoder-only, 125M–1.3B; 12–24 blocks.
- OuterOPT: parameter averaging; InnerOPT: AdamW.
- Data: The Pile (16 sources); MC4 (EN/IT/ZH/SR/MS/SW/UR/LA).
- Baselines: STD [Devlin'19; Conneau'20; Xue'21]; ACT [Chen'23].
- Metrics: perplexity (in/OOD), downstream (MNLI, RACE, STSB, SST-2), plasticity, stability.

Baseline References (for Slide 10)

- STD = Standard distributed pretraining approaches aligned with Devlin et al., 2019 (BERT); Conneau et al., 2020 (XLM/XLM-R); Xue et al., 2021 (mT5).
- ACT = Pre-training with active forgetting, Chen et al., 2023.

See full citations on References (Slide 23).

Results: Robustness to Heterogeneity (RQ1)

- The results indicate stabilized activation and parameter norms versus STD under identical local hyperparameters.
- Regularization effects are consistent with OuterOPT noise injection and meta-learning characteristics.

[Figure 3: norms over steps placeholder]

Limitations and Caveats

Following the RQ1 robustness results, we highlight practical constraints and design implications.

- SPEC requires a final global embedding for practical deployment.
- Effective continued pretraining design is needed to optimize OOD embedding performance.
- Further exploration is warranted for zero-shot embedding transfer, vocabulary matching, and model stitching.

Results: Efficiency (RQ2)

- Experiments reveal large communication cuts versus DDP/FSDP; GLOB aligns with Local SGD behavior.
- TRIM yields 2%–78% vocabulary shrinkage on The Pile and 8%–32% on MC4.
- SPEC eliminates embedding communication and adds 13–30% further savings (multi-domain) and 34% (multilingual).
- At 1B scale, SPEC-OPT attains ~714× communication reduction and ~24% memory reduction.

[Cost table/plot placeholder]

Results: Generalization (RQ3)

- With randomly initialized embeddings after continued pretraining, average perplexity improvements of ~15–20% are observed across The Pile and MC4.
- SPEC-OPT excels on high UNIGRAM-CE sources (e.g., EP, DM, GH) and OOD UI.
- TRIM performs comparably to GLOB despite smaller embeddings; DEPT variants win most comparisons.

[Tables 3 & 4 summary placeholder]

Results: Pretrained Embeddings (Continued Phase)

- On The Pile, GLOB/TRIM embeddings outperform STD in many cases (average +5.5%).
- On MC4, gains appear for in-distribution while OOD is mixed—sampling and tokenizer choices matter.
- Transformer superiority from DEPT persists when starting from random embeddings.

[Tables 5 & 6 placeholder]

Results: Downstream Tasks

- Consistent gains on MNLI, RACE, STSB, SST-2, with TRIM strongest overall.
- DEPT outperforms ACT in most tasks while avoiding costly reinitialization cycles.
- Improvements align with regularization via aggregation noise.

[Table 7 placeholder]

Results: Plasticity (RQ4)

- Faster adaptation and lower final perplexity on new domains/languages (e.g., HI, DE; SW as smallest-vocab case).
- Stable convergence is observed across settings.

[Figure 4 adaptation curves placeholder]

Continued Pretraining for SPEC

- After DEPT, a global embedding is created via multi-phase adaptive pretraining.
- Embeddings start from random initialization with a tokenizer tailored to the corpus (e.g., The Pile / MC4).
- This stage ensures comparability with STD and evaluates transformer body quality.

When to Use Each Variant

GLOB

- Fits centralized/federated settings with a known global vocabulary.
- Provides lower communication than per-step synchronization with simple deployment.

TRIM

- Targets memory/communication sensitivity while retaining GLOB’s assumptions.
- Is suitable for large global vocabularies with tied weights and modest perplexity impact.

SPEC

- Addresses unknown/private data with per-source tokenizers.
- Minimizes communication; is vocabulary/length agnostic.
- Requires a final embedding generation step for deployment.

Qualitative Example: Per-Source Tokenization

- Examples (conceptual): DM (math) and GH (code) receive specialized tokenizers, while EP (parliament) uses a different subword mix.
- Viewers should note how differing vocabularies reduce dilution and better match domain-specific patterns, enabling lower perplexity and faster adaptation reported in the results.

[Sample token splits for DM: "int_x^y f(t)dt" → [int, _, x, ^, y, ...]; GH: "for(i=0;i

Related Work

- Multilingual pretraining (mBERT, XLM, mT5) with sampling/vocabulary strategies.
- Language clustering, knowledge transfer, and domain adaptation face limitations on low-resource data and stability.
- Active forgetting improves plasticity but is compute-inefficient relative to DEPT.

References (Selected)

- Devlin'19 = Devlin et al., 2019 (BERT); Conneau'20 = Conneau et al., 2020 (XLM-R); Xue'21 = Xue et al., 2021 (mT5); Rust'21 = Rust et al., 2021.
- McMahan'17 = McMahan et al., 2017 (FedAvg); Stich'19 = Stich, 2019 (Local SGD); Gururangan'20 = Gururangan et al., 2020.
- Zhang'22 = Zhang et al., 2022 (OPT); Chowdhery'23 = Chowdhery et al., 2023 (PaLM); Lin'20 = Lin et al., 2020.
- Chen'23 = Chen et al., 2023 (Active Forgetting); Xu'24 = Xu et al., 2024; Tao'24 = Tao et al., 2024.

[Additional figures/tables placeholder]

Takeaway

Decoupling embeddings from the transformer body enables vocabulary-agnostic, communication-efficient pretraining that improves generalization and scales to billion-parameter models.